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MH1200 Linear Algebra I

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1. Review of Determinants

Minor Matrix Matrix that results from removing a row/column
Minor Determinant of a Minor Matrix
Cofactor A Minor times the sign $(-1)^{i+j}$

2. Six Elementary Properties of Determinants

1. If a matrix A has a row of zeroes or a column of zeroes, then $\det(A) = 0$.
2. $\det(A^T) = \det(A)$.
3. Determinants are linear functions with respect to individual rows, and also individual columns.
4. If a matrix A has a repeated row or a repeated column, then $\det(A) = 0$.
5. If we obtain a matrix A' from a matrix A by swapping two rows, or swapping two columns, then $\det(A') = -\det(A)$.
6. The determinant of an upper-triangular matrix is just the product of the diagonal entries.